Universität Konstanz

Abstraction Mechanisms for Virtual Avatars

Non-Photorealistic Rendering across Avatar Platforms
and its Effect on Perceived Trust

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Master's Thesis Defense

First reviewer: Prof. Dr. Oliver Deussen
Second reviewer: Prof. Dr. Tiare Feuchtner

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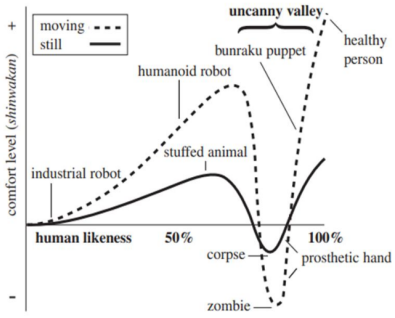
The problem

Realism does not buy trust

- Virtual humans now advise, teach and sell. How they look changes how much people believe them.
- Human likeness raises trust, **but not linearly**. Almost-human characters fall into the **uncanny valley**.
- **Non-Photorealistic Rendering (NPR)** moves an avatar back down the realism axis on purpose.

1. Avatar platforms expose very unequal rendering access.

2. Nobody has measured what the abstraction does to trust.



The uncanny valley (Mori, 1970)

Research questions

RQ1 Technical transfer

How can NPR techniques be adapted to avatar platforms with different rendering architectures, and what does each platform allow or forbid?

RQ2 Technique selection

Which technique or combination best balances visual abstraction against facial legibility?

RQ3 Perception

How does NPR stylisation affect perceived eeriness and trustworthiness, and can a stylised avatar reach a photorealistic avatar and a real human speaker?

RQ1 and RQ2 are answered by the system. RQ3 is answered by the user study, which is where this talk spends most of its time.

What was built

Eight techniques, three platforms, one shader system

Version	Limitation it answers
V1 Silhouette outline	no shape cue at all
V2 Toon shading	continuous, unreadable lighting
V2.2 X-Toon	no control over abstraction level
V3 Normal edges	no interior structure
V4 Sobel	misses colour boundaries
V4.2 Gaussian Sobel	Sobel is noisy on photo texture
V5 Hierarchical	one cue is not enough
V6 Kuwahara	edges only, no painterly surface

Each version was motivated by a concrete failure of the one before it. All are switchable at runtime, without recompiling.

Platform	Access
Mixamo	full
Avaturn	full, photo texture
Meta SDK	**one fragment hook**

Unity 2022.3 LTS, URP.
Meta Quest 3 target.
No extra render passes on the SDK.

RQ1 answer

What transfers between pipelines, and what does not

Transfers with retuning

- Every technique ran on all three platforms.
- Identical parameter values produced **visibly different** output per host.

Needs a substitute signal

- Meta SDK gives one fragment hook, no depth or normal buffer, no second pass.
- Hierarchical layer: camera-distance derivative **instead of** a depth texture.
- Kuwahara: loses its Gaussian smoothing entirely.

The dividing line

Not the technique's complexity, but whether it needs **new parameter values** or a **new input signal**.

New values: always possible.
New signal: only where the pipeline is open.

Avaturn was chosen for the user study because it is the only platform with all three of: open URP shader access, full blendshape animation, and a photorealistic face rebuilt from photos of the real speaker.

The stimulus

Same person, same animation, same voice

C1 — Real

Video of the human speaker.

Ground-truth reference.

The Avaturn avatar was reconstructed from photographs of this same person.

C2 — Photorealistic



Standard PBR, unmodified texture

C3 — NPR



Outline + X-Toon + Sobel, preprocessed head texture

Takeaway. C2 and C3 share identity, geometry, animation and voice. Only the rendering changes. That is the critical comparison.

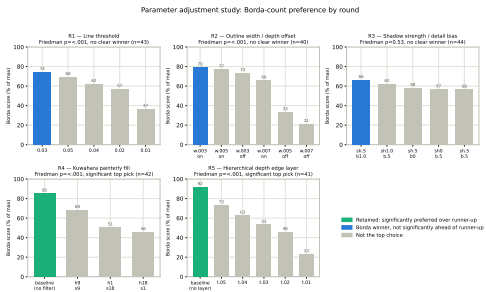
Parameter adjustment study

The composite was chosen by people, not by eye

Design. 45 participants ranked rendered variants by **perceived trustworthiness**. Five rounds, one parameter group each. Borda count, Friedman omnibus, Wilcoxon on the top pair.

Round	Winner	Significant?
R1 inner threshold	0.03	no ($p = .31$)
R2 outline width	0.003	no ($p = .82$)
R3 shadow / bias	0.5 / 1.0	no ($p = .53$)
R4 Kuwahara	**none**	**yes** ($p = .003$)
R5 Hierarchical	**none**	**yes** ($p < .001$)

R1 and R3 were settled by a separate visual appeal question.



Borda score per round, % of maximum

Takeaway. Participants significantly *preferred no post-processing*. Kuwahara and the hierarchical layer were both removed. RQ2's answer is a three-part composite plus head texture preprocessing.

Main user study

Within-subjects, video based

Task. Read a decision vignette, watch an advisor recommend one option, then rate the advisor. Three times, once per condition.

Scenarios. Career choice, travel route, learning resource. Each targets a different trust dimension. Recommended option and discussion order alternate.

Counterbalancing. 3×3 Latin square, three study versions.

Sample. 52 enrolled, 11 excluded on gating checks, $n = 41$ complete.
Age $M = 27.6$ ($SD = 5.4$); 23 women, 17 men, 1 non-binary.

Measures per video block

Subscale	Source	α
Likeability	Godspeed	.94
Eeriness	Ho & MacDorman	.87
Ability	Mayer et al.	.96
Benevolence	Mayer et al.	.95
Integrity	Mayer et al.	.92

After all three blocks

> Forced ranking: who would you consult, who is most trustworthy, who is least unsettling.

All subscales exceeded $\alpha = .70$ overall and within every condition.

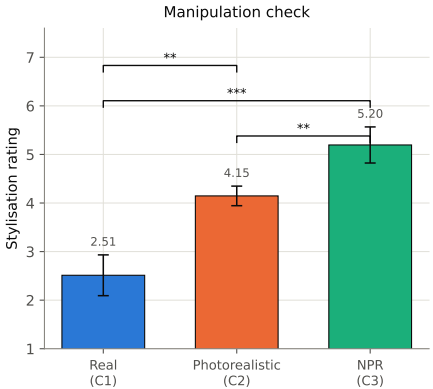
Four hypotheses

	Prediction	Test
H1	NPR (C3) is rated *less eerie* than the photorealistic avatar (C2).	RM-ANOVA, Holm pairwise
H2	NPR (C3) does *not* cost trust relative to C2.	Pairwise **plus** TOST equivalence, $d = 0.3$ bound
H3	Eeriness lowers trust and likeability raises it, within each participant.	Person-centred multilevel regression
H4	NPR (C3) does *not* cost trust relative to the real speaker (C1).	Same as H2, against C1

H2 and H4 predict a **null**. A non-significant p is not evidence of equivalence, so both also carry a pre-specified TOST equivalence test.

Manipulation check

Did participants actually see a difference?



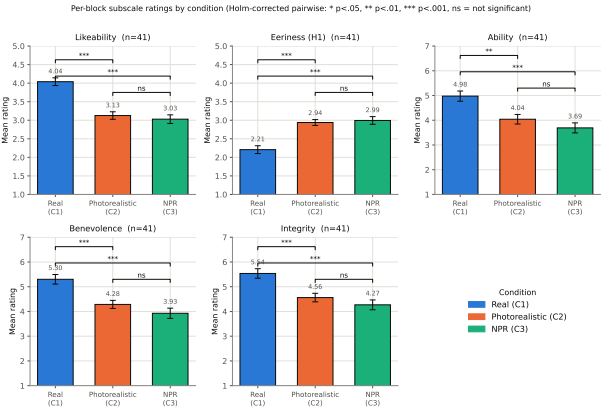
Part H asked how stylised each advisor looked, from 1 (completely realistic) to 7 (very stylised).

Omnibus	$F(2, 80) = 15.45$, $p < .001$, $\eta_p^2 = .28$
C1 vs C2	$p = .001$, $g = 0.87$
C1 vs C3	$p < .001$, $g = 1.23$
C2 vs C3	$p = .004$, $g = 0.55$

Ratings rise monotonically from Real to Photorealistic to NPR. All three conditions separate.

Takeaway. The manipulation worked. No null result below can be blamed on participants not noticing the rendering.

All five subscales at once



The real speaker separates from both avatars on every subscale.

The two avatars separate from each other on *none* of them.

H1 — Does NPR reduce eeriness?

Eeriness	C1 vs C2	C1 vs C3	C2 vs C3
p	$< .001$	$< .001$	.66
g	0.98	0.98	0.07

Omnibus: $F(2, 80) = 20.39$, $p < .001$, $\eta_p^2 = .34$

- Eeriness rose from C1 to C2 to C3, the **opposite** of the prediction.
- The real speaker differed from both avatars.
- The two avatars were **statistically indistinguishable**. Effect size $g = 0.07$ is essentially zero.

H1 NOT SUPPORTED

Participants *saw* the stylisation, as the manipulation check confirmed. They just did not find it *less eerie*.

Stylisation and eeriness turn out to be separate effects here.

H2 — Does NPR cost trust against the photorealistic avatar?

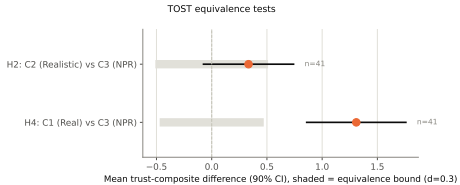
Step 1. Pairwise comparisons, C2 vs C3

	p	*g*
Ability	.21	0.20
Benevolence	.18	0.19
Integrity	.27	0.16

Step 2. TOST equivalence, trust composite

	Diff.	Bound	TOST *p*
C2 vs C3	0.33	±0.51	.240

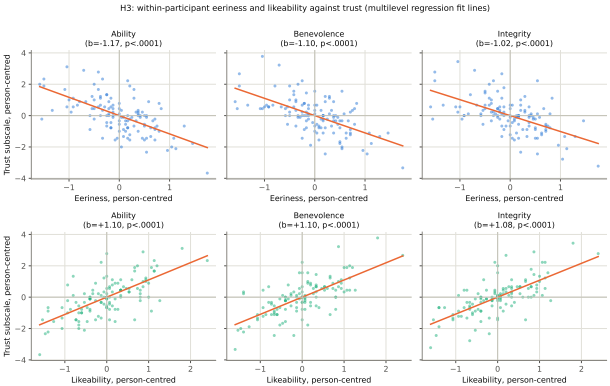
Realised power near 0.20 at $n = 41$.



H2 INCONCLUSIVE No difference detected, but equivalence not established either.

Roughly **three times** the sample is needed to separate the two.

H3 — Eeriness, likeability and trust



	Eeriness	Likeability
Ability	-1.17	$+1.10$
Benevolence	-1.10	$+1.10$
Integrity	-1.02	$+1.08$

Fixed-effect slopes, all $p < .001$

H3 CONFIRMED

One point more eeriness costs about one point of trust, on every dimension. Likeability lifts it by the same amount.

Caveat: the two correlate at $r = -.72$ within participants and cannot be separated cleanly here.

Takeaway. The route from appearance to trust is open. The composite never travelled it.

H4 — Does NPR reach the real human speaker?

Pairwise, C1 vs C3. Ability, benevolence and integrity were **all significantly higher** for the real speaker ($p < .001$, $g = 0.75$ to 0.80).

TOST equivalence, trust composite

	Diff.	Bound	TOST p
C1 vs C3	1.31	$\pm$0.47	.998

The difference is nearly **three times** the equivalence bound. This is not a marginal shortfall.

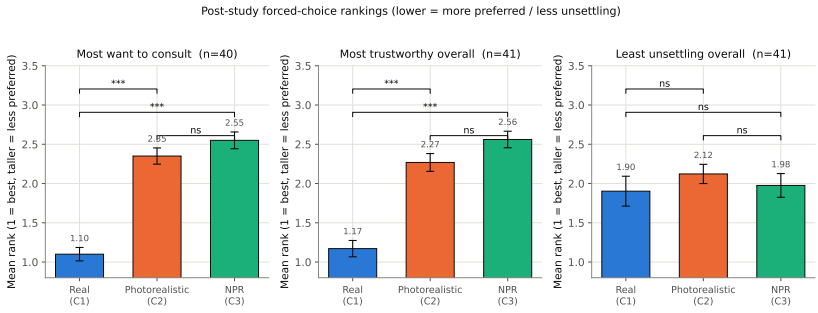
H4 NOT SUPPORTED

Two explanations stay open and this design cannot separate them:

1. The uncanny valley predicts exactly this gap.
2. Some participants may have recognised the speaker, who is the researcher.

The behavioural measure

Forced ranking after all three videos



Question	Friedman	C1 vs C2/C3	C2 vs C3
Most want to consult	$\chi^2_F(2) = 49.40, p < .001$	$p < .001$	$p = .21$
Most trustworthy	$\chi^2_F(2) = 44.05, p < .001$	$p < .001$	$p = .07$
Least unsettling	$\chi^2_F(2) = 1.02, p = .60$	$p = .97$	$p = .97$

Canales et al. found ratings and behaviour *diverge*. Here they **converged**. C3 attracted no more behavioural endorsement than C2.

Robustness checks

Three ways the null could have been an artefact, and was not

Order effects

Video position had no significant effect on any subscale. Likeability came closest at $p = .054$.

Study version groups differed on likeability and eeriness, but on no trust subscale. That comparison is between subjects.

Suspicion

24 of 41 named realism, stylisation or trust when asked what the study was about.

Re-running H1 and H2 on the remaining 17 did not change the pattern.

Participants answered on their own impressions, not on a guessed hypothesis.

Secondary measures

Agreement with the recommendation did not differ by condition, $F(2, 80) = 2.10$, $p = .13$.

Dispositional trust did not predict the trust composite, $r(39) = -.10$, $p = .52$.

Preference shift was rare everywhere: 4, 5 and 1 of 41.

Discussion

Two patterns that need explaining

Pattern 1

The real speaker sits far above both avatars, on every subscale and both significant rankings.

Pattern 2

The two avatars looked different to participants, yet scored identically on everything that mattered.

And one constraint on any explanation:

Eeriness and likeability *did* predict trust (H3). The road was open.

The composite changed what people saw without travelling it.

Why eeriness did not fall

Two mismatches inside the composite

1. Material versus shape

- The composite stylises the **material** and leaves the **shape** untouched.
- Zell et al.: material drives eeriness, so this should have helped.
- But the same work reports eeriness **rises** when material and shape disagree in stylisation level.
- C3 is exactly that case: a stylised surface on a photogrammetric face.

The abstraction may have removed one source of eeriness while adding another.

2. Appearance versus performance

- Both avatars inherit the **same** recorded human voice, gaze and lip animation.
- C3 therefore pairs a fully human performance with a visibly abstracted face.
- Cihodaru-Ştefanache & Podina: uncanniness arises from features perceived at **inconsistent** levels of anthropomorphism.
- A shading operator changes appearance only. It could only widen that gap.

Three more explanations on the table

The comparison set

Once a real person anchors the top of the scale, the only judgement left is *how far below*.

Both avatars were recognisably CG. A rendering difference competes against a categorical one.

C3 followed C1 directly in two versions of three. C2 never did.

Generative AI expectations

One participant asked afterwards *which AI system generated the videos*.

Nothing in the materials ruled that reading out.

Scepticism towards machine-made media depresses *both* avatars equally.

The video setting

Flat screen, prerecorded, no interaction, no shared space.

Zibrek et al. found render style significantly moved presence in VR.

What is too weak in a browser window may register once the avatar shares your room.

Limitations

Session length

Both studies had to fit 10 to 15 minutes, and every measure was repeated three times. That capped how many measures and conditions were affordable.

One configuration only

One composite, one avatar identity, one platform, three scenarios. The findings describe **this configuration**, not NPR as a category.

Sample size

The submission deadline set the collection window, which fell in the semester break. $n = 41$ met the power requirement for *detecting* a difference, not the far larger one *equivalence* demands.

Counterbalancing

Three conditions need six orders for a full design. That needs roughly twice the sample, so three orders were used.

The parameter study also tested only the highest-impact groups, so the composite is suitable rather than globally optimal.

Conclusion

What the three questions ended up answering

RQ1 Adaptation across platforms

All eight techniques ran on all three platforms. The dividing line is whether an operator needs **new parameter values**, which is always possible, or a **new input signal**, which only an open pipeline provides.

RQ2 Best balance of abstraction and legibility

No technique reached it by rendering alone on a photogrammetric avatar. The answer is a **three-part composite**, outline plus X-Toon plus Sobel, **paired with head texture preprocessing**.

RQ3 Effect on eeriness and trust

H1 not supported, H2 inconclusive, **H3 confirmed**, H4 not supported. The composite did not lower eeriness, and it also carried **no observed trust penalty**.

A developer choosing the stylised version need not assume a disadvantage.
The real distance is not between two render styles. It is to the actual human.

Future work

Settle the open question

- Repeat C2 versus C3 with roughly **120 participants**. Inside the bound confirms no trust cost, outside exposes one too small to see here.

Test the competing explanations

- Extend abstraction to the **geometry**, not just the material.
- Drop the real speaker and let the two avatars be compared alone.
- State up front that the videos are **not AI-generated**.

Replace the point with a curve

- Sweep abstraction strength continuously instead of testing one setting. That traces how eeriness and trust move together and can reveal an optimum between the extremes.

Restore what video removed

- Interaction and shared space, in VR.

Widen the sample of avatars

- Multiple identities and platforms, to separate a general effect from one specific to this face.

Thank you

Questions

Eight NPR techniques, three avatar platforms, two studies, 86 participants.

Stylisation was clearly seen, cost no measurable trust,
and did not close the gap to a real human.

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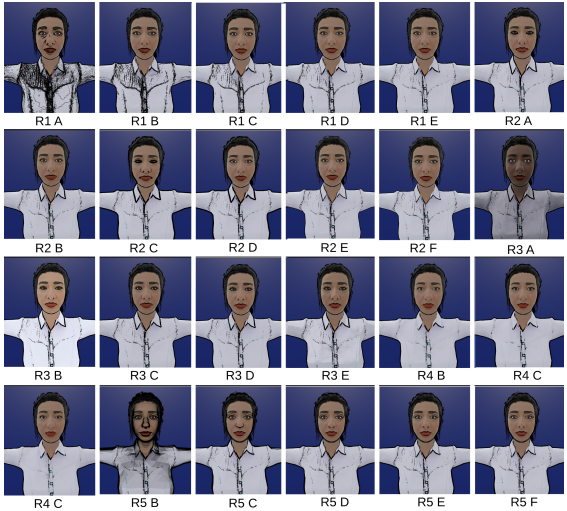
Backup slides

Backup — Visual quality assessment of all eight techniques

Technique	Facial	Temporal	Artefacts	Abstraction	Candidate
V1 Outline	yes	yes	after tuning	**not alone**	yes
V2 Toon	yes	yes	yes	**no interior lines**	—
V2.2 X-Toon	yes	yes	yes	manual mode chosen	yes
V3 Normal edge	**landmarks missed**	yes	yes	yes	—
V4 Sobel	after tuning	after tuning	after preprocessing	yes	yes
V4.2 Gauss Sobel	**no gain over V4**	after tuning	yes	yes	—
V5 Hierarchical	**segments unclear**	yes	**nose bridge artefact**	yes	yes
V6 Kuwahara	**fails at large radius**	yes	yes	yes	yes

Five of the eight became candidates for the composite. The parameter adjustment study then removed V5 and V6, leaving V1, V2.2 and V4.

Backup — All variants shown in the parameter adjustment study

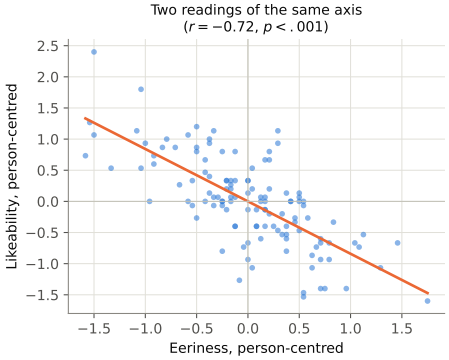


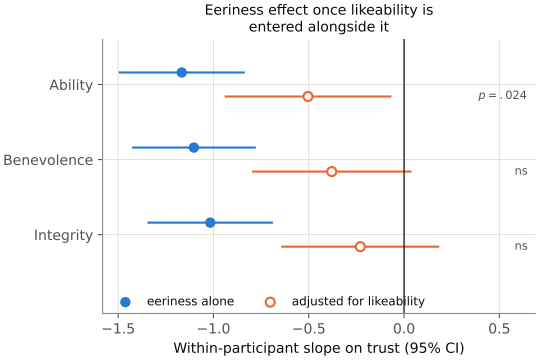
Backup — Final composite parameters

Parameter	Value	Source
Outline width	0.003	R2 Borda winner
Depth offset	On	R2 Borda winner
Inner line threshold	0.05	R1 appeal tiebreaker
Shadow strength	0.0	R3 appeal tiebreaker
Detail bias	0.5	R3 appeal tiebreaker
Kuwahara filter	**Off**	R4 excluded ($p = .003$)
Hierarchical layer	**Off**	R5 excluded ($p < .001$)

Three of the seven values came from a separate visual appeal question, since no configuration in those rounds was statistically ahead of its runner-up.

Backup — Eeriness against likeability





Within participants the two correlate at $r = -.72$, sharing 52% of their variance. Entering both in one model absorbs most of the eeriness effect: it survives on ability ($b = -0.50$, $p = .024$) but not benevolence ($p = .075$) or integrity ($p = .273$), while likeability holds on all three.

Backup — Full statistics table

Subscale	Omnibus	C1 vs C2	C1 vs C3	C2 vs C3
Likeability	$F(2, 80) = 26.85$, $p < .001$, $\eta_p^2 = .40$	$p < .001$, $g = 0.92$	$p < .001$, $g = 1.02$	$p = .54$, $g = 0.08$
Eeriness	$F(2, 80) = 20.39$, $p < .001$, $\eta_p^2 = .34$	$p < .001$, $g = 0.98$	$p < .001$, $g = 0.98$	$p = .66$, $g = 0.07$
Ability	$F(2, 80) = 10.91$, $p < .001$, $\eta_p^2 = .21$	$p = .004$, $g = 0.53$	$p < .001$, $g = 0.75$	$p = .21$, $g = 0.20$
Benevolence	$F(2, 80) = 14.31$, $p < .001$, $\eta_p^2 = .26$	$p < .001$, $g = 0.59$	$p < .001$, $g = 0.79$	$p = .18$, $g = 0.19$
Integrity	$F(2, 80) = 12.47$, $p < .001$, $\eta_p^2 = .24$	$p = .001$, $g = 0.66$	$p < .001$, $g = 0.80$	$p = .27$, $g = 0.16$

Order and version checks

Subscale	Position $F(2, 80)$	Position p	Study version $F(2, 38)$	Study version p
Likeability	3.04	.054	**7.07**	**.002**
Eeriness	2.47	.091	**5.57**	**.008**
Ability	1.32	.274	1.23	.304
Benevolence	2.10	.130	1.26	.296
Integrity	1.72	.186	2.04	.144

Intraclass correlations, H3 models

Eeriness models: .48 to .57.
Likeability models: .57 to .63.

Half or more of the variance in each trust subscale sits *between* participants rather than within them, which is consistent with large individual differences in general trustingness.